

Vishing Shield: A Conceptual Framework and Detection Model to Mitigate Vishing Attacks in Mobile Financial Services (MFS)

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Abstract—The increasing adoption of Mobile Financial Services (MFS) by banks and financial institutions has significantly enhanced digital payments, providing consumers with added convenience and easy access. However, the rapid growth has also raised serious security concerns, especially regarding vishing attacks. Vishing, also known as voice phishing, is a technique used by scammers to deceive victims through fake calls to obtain sensitive bank information. Thieves exploit the ability of call paths being masked, making them difficult for law enforcement to trace. Research has established that vishing attacks remain unreported due to many victims blaming themselves for being tricked. Compared to other traditional methods like email spoofing, vishing still lacks comprehensive documentation despite its increasing prevalence. This paper examines the major drivers of vishing attacks and identifies effective countermeasures. A cross-sectional survey was conducted among 100 mobile phone users using a structured questionnaire to collect data. Among them, 82% of users who are acquainted with social engineering attacks, 44.9% had been the victim of vishing attacks, and 35.6% experienced vishing attacks due to MFS services to a very great extent. Based on these results, preventive methods were suggested, and a coding scheme and a vishing attack detection model were designed to mitigate risks in the MFS ecosystem. The model presented here aims to strengthen security controls, promote user awareness, and counter the impacts of vishing fraud.

Index Terms—Cyber Security, Vishing Attacks, Social Engineering Attacks, Psychological Factors, Technical Factors, Mobile Financial Services

I. INTRODUCTION

With the advent of the digital age, internet users, governments, and service providers, especially financial institutions, are increasingly exposed to social engineering attacks. Of these, vishing frauds have registered a sharp growth in recent times, threatening the world's cybersecurity severely [1]. Vishing is an advanced social engineering tactic where fraudsters employ telephone calls to trick victims into revealing confidential financial or personal details [2]. The majority of the victims are unaware that attackers employ caller ID spoofing and advanced automated technology to make the attacks appear realistic. Phishing is no longer effective, and, therefore, cybercrime gangs have shifted to vishing to intercept financial data, passwords, and other confidential information. This shift has significantly altered the data security risk scenario [3]. As digital transactions have grown, so has the risk of cyber fraud, and therefore, security measures have

become more necessary than ever. Bangladesh has emerged as a global leader in mobile financial services (MFS), as an alternative to banking, as described in the Bangladesh Bank Policy Paper [4]. While these technological advancements have facilitated financial transactions, they have also opened new avenues for cybercrime activities. Cybercriminals now target financial service organizations instead of traditional banks to gain unauthorized access to banking systems. With most of the banks in Bangladesh providing enhanced mobile-based services, customers are now able to manage loans, deposits, and transactions through smartphones. Studies, such as that of M. N. Nabi et al. [5], point out that social engineering remains a dominant cybersecurity problem in Bangladesh. There have been many reported cases where individuals were fooled into sharing confidential information, e.g., SIM PINs or mobile banking IDs, leading to unauthorized transactions. The increasing rate of vishing attacks on Bangladeshi mobile platforms proves the urgent need for effective countermeasures. To manage these security flaws, Bangladesh needs to enhance international collaboration, increase technological expertise, and enhance public awareness related to digital fraud. If not dealt with proactively, mobile transactions within the country will remain highly vulnerable to cybercrime activities, which may affect the trust and security of millions of users. Bangladesh is leading the world in the use of mobile payments as an alternative to conventional banking, according to M.S.R. Bhuiyan et al. [6]. These developments are viewed as new payment methods for internet businesses that offer simpler money access. In order to get access to bank systems, cybercriminals now target financial services firms rather than banks [5]. The majority of Bangladeshi banks are working to make all of their services available on mobile platforms [4]. Customers may use their phones to process loans as well as make deposits and withdrawals [3]. The study by Foozy et al. [7] covers the present trend of mobile device phishing detection and identifies important criteria to enhance mobile device phishing detection systems. As a result of their investigation, a general taxonomy for phishing detection on mobile devices is proposed. The study, as per Keith et al. [8], Vishing assaults tended to utilize authority (A), social proof (S), and diversion (D) more frequently than they did liking, likeness and deceit (L). The bulk of vishing attacks used these four influential techniques. Cybercriminals and other hostile actors are now aware of the advantages of artificial intelligence, according to Qamar et al. [9], and they attempt to incorporate it into their nefarious actions. According to Parthy and Rajendran [10], social engineering is the most straightforward method of taking control of a system since it can be executed quickly and with little technical know-how, targets people, and exploits a variety of psychological flaws in human cognition. These assaults are frequently used to target businesses since people are their weakest link and are easily tricked and influenced. In 2014, Yeboah-Boateng et al.'s study [11] carefully questioned twenty (20) end users on their knowledge, attitudes, and actions when faced with phishing attack scenarios. The findings indicate that males are more at ease

and trust online than women, making them more vulnerable to phishing scams. Fascinatingly, 55% occasionally scrutinized communications they received for threats, compared to 35% who never or hardly ever did so. Additionally, there is a shortage of the theoretical foundation required to develop effective countermeasures, as well as a lack of understanding of the context of the situation and features of voice phishing [12].

The main goal of this research is to provide a model for mobile users to recognize phishing attacks. This paper's particular goals are to identify vishing attack contributors, create a mobile vishing attack detection model, and evaluate the vishing attack detection model. This study seeks to develop a model for detecting phishing attacks to help mobile phone users better recognize social engineering assaults. The model will be used as a teaching tool to raise awareness of phishing attacks by the government and financial institutions.

II. METHODOLOGY

A structured 100-user survey was conducted, and its observed indicators directly guided the variable and rule selection. This section presents the steps followed to study vishing attacks in MFS systems. It includes the attack framework, key control variables, decision-making rules, coding logic, and user feedback analysis. Each part builds toward developing a rule-based model for vishing detection. Vishing is a type of phishing assault in which social engineers use phone conversations to trick people into giving critical information that they may later use against themselves [12], [13]. For a phisher, a typical vishing attack is obtaining the authentication details for a victim's MFS account, which the attacker has compromised, and using that information for other purposes. Figure 1 depicts the conceptual framework of a vishing attack with the key components of a social engineering-based attack that satisfy all the study's objectives.

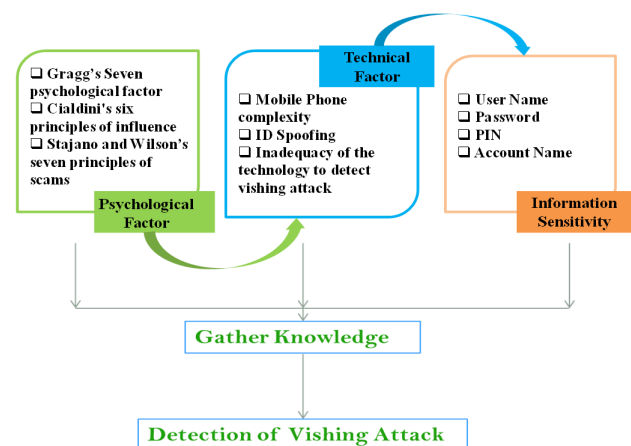


Fig. 1. Conceptual framework for detecting vishing attacks.

This framework influences a previously established ontological model for structuring the components of social engineering attacks, and a more systematic analysis will be

ensured. The ability to represent temporal data, capturing the flow of attacks, and the timing of attacks is the key advantage of this model. By utilizing this model, we can gain a more comprehensive understanding of the nature of social engineering attacks. This structure includes the categories, which are goal determination, attack planning, execution, and completion. The Model categorizes different aspects of social engineering attacks by integrating the ontological elements, which enhances the study of their dynamics and progression. This framework also provides some valuable insights, such as how attackers manipulate their targets and how to execute their strategies. Figure 2 illustrates the structure of this social engineering attack framework, its components, and their interactions. This enhanced model is more robust than previous models, and it contributes to cybersecurity research and defense strategies.

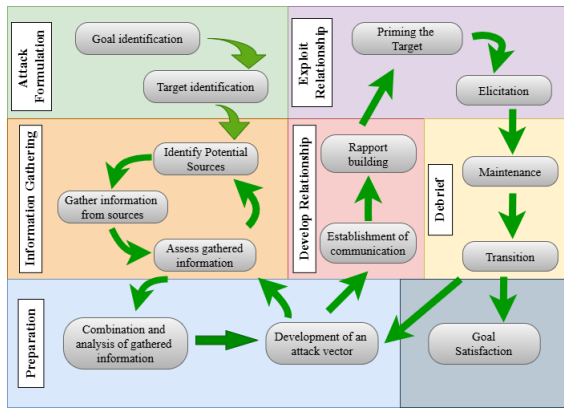


Fig. 2. Social engineering attack framework.

The Social Engineering Attack Detection Model (SEAD), developed by Bezuidenhout, Mouton, and Venter in August 2010 [13], was specifically designed for use in call center environments. Its primary purpose is to aid call center employees in identifying potential social engineering attacks. These attacks will occur when attackers manipulate a call center worker into disclosing sensitive information. SEAD is planned for providing a quick and efficient evaluation of whether a requester is engaging in misleading tactics. By implementing this model call center personnel can make informed decisions about information disclosure. This model serves like a decision-making aid that ensures that sensitive data is only shared with individuals who have the proper authorization. The ability to analyze various cues and callers' behavior is the effectiveness of SEAD which reduces the risk of social engineering attacks. The model will be introduced by the thinking of enhancing security measures within call centers by preparing employees to recognize and respond appropriately to suspicious request by providing them necessary tools. A visual representation of SEAD will be given below in Figure 3 which illustrates its framework and within a call center setting for detecting social engineering threats.

We aim to develop a model based on the data gathered from surveys and key informants. Vishing attacks are

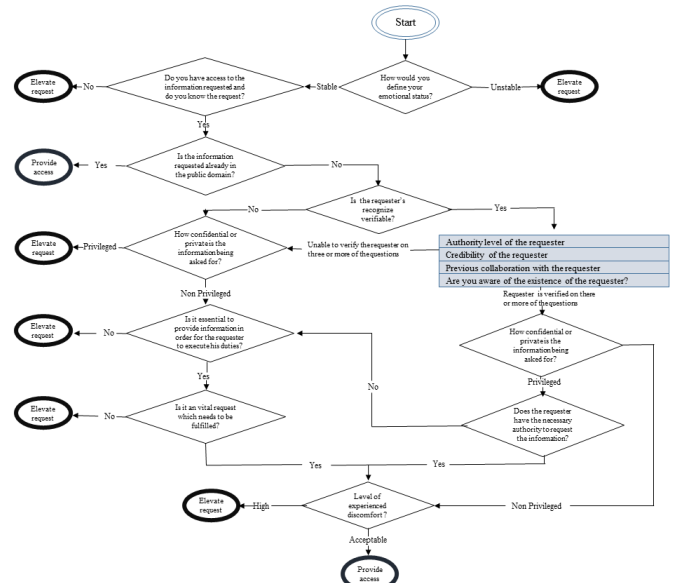


Fig. 3. Social engineering attack detection model.

mainly driven by psychological factors, technology-related issues, and the sensitivity of the information involved. The two most important factors in preventing vishing attacks are increasing psychological awareness and providing computer security training. This study presents a conceptual, rule-based model, and due to limited dataset size, extensive computational validation is reserved for future work. To develop the rule-based model, we have followed the following steps:

A. Set the Control Variables

Based on the data and analysis, we found that technology and people formed two major vulnerabilities that could be easily exploited by social engineers on mobile platforms. We added information sensitivity as the key factor in the detection of vishing attacks. The SEADM, developed by Bezuidenhout et al. [13] used only the psychological and information sensitivity vulnerabilities in the detection of social engineering attacks. Besides this vulnerability, we added a technical vulnerability for the detection of vishing attacks. In this study, the variables are categorized as follows:

- Independent variables – psychological factors, technical factors, and information sensitivity;
- Intermediate variable – knowledge;
- Dependent variable – vishing detection.

These variables were chosen based on patterns identified directly from the survey responses.

B. Decision Making Rule

The following decision-making rules were developed using IF-THEN logic, aiding the identification of vishing attacks based on behavioral and technical indicators. These rules are categorized into distinct groups, reflecting the typical flow and structure of vishing attack strategies.

TABLE I
GROUPING OF RULES FOR DECISION-MAKING IN VISHING ATTACK
DETECTION

Group	Rule
ID Spoofing	Phone Number
Strong Affect	Emotion
Overloading	Script
Sensitivity / Information	Request
Security	Level
Complexity	Level

C. Coding Scheme

User input features: Phone Number, Emotion, Level, Script,
Request Vishing $\leftarrow$ True/False decision

Group: ID Spoofing

Rule: Phone Number

Phone Number = Not recognized **or** Phone Number = hidden
ID_Spoofing $\leftarrow$ likely

Group: Complexity

Rule: Level

Level = complex Level $\leftarrow$ Exploited

Group: Overloading

Rule: Script

Script = Not clear **or** Script = Not clarified

Overloading $\leftarrow$ triggered

Group: Strong Affect

Rule: Emotion

Emotion = Unstable

Strong_Affect $\leftarrow$ triggered

Group: Information

Rule: Request

Request = password **or** Request = pin **or** Request = username

Information $\leftarrow$ Sensitive

Group: Vishing

Strong_Affect **and** Overloading **and** Information = Sensitive
Vishing $\leftarrow$ True

Now, the overall process of the algorithm can be depicted
as shown in Figure 4.

D. Users Evolution

An experimental analysis is conducted based on real users' feedback. The feedback of the users is processed under some popular mathematical operations to reach the outcome of the experiment. Users generate their feedback on the basis of their experience by using MFS on their mobile phones and also the users will use only one independent variable to evaluate the survey. The survey that was used in this retrospective investigation was thoroughly examined to make sure that the data collected was displayed in a way that made sense, preferably with the help of percentages and diagrams. In order to further discuss the elements that lead to vishing attacks, the information produced was statistically examined. Here the

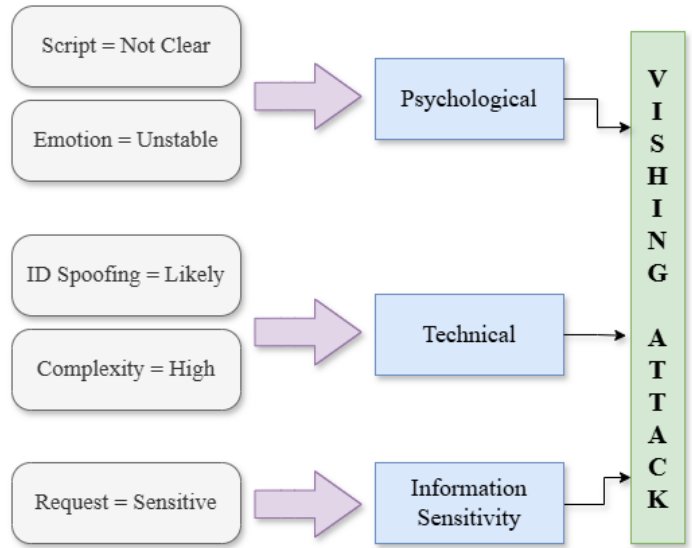


Fig. 4. Vishing attack detection model.

data was conducted via Google form and the total number of participants was around a hundred.

III. RESULTS AND ANALYSIS

Analysis and Findings are processed in two ways: Findings from users' feedback and Analysis of Dependent Variables. The data assessment, discussion, model building, and growth are all covered in this part. The results are given by the methodology that has been explained, the conceptual framework, and the study goals in order to address the research issues. The study of the data that was gathered produced the findings in this part. Hundred (100) of the surveys that were given to mobile users with different origins and backgrounds were filled out and returned. These answers are deemed suitable to help with drawing conclusions and offering advice.

A. Types of Social Engineering Attacks

Attacks using social engineering can be carried out anywhere there is a chance of human contact. The study sought to find out which social engineering techniques are the most common forms of digital SE assaults. Maximum of the respondents 37.1% had been victims of the phishing attack, 44.9% had been the victim of vishing attacks, 11.2% had been the victim of pretexting and 6.8% had been victims of other types of social engineering attacks. That comparison is graphically shown in Figure 5. This study concludes that the vishing attack is at in high of social engineering attacks.

B. Exposure to Vishing Attacks

The goal of the research was to determine the number of vishing assaults that had occurred. A maximum of the respondents, 44.9% had been victims of SE attacks, with 4.4% experiencing vishing attacks to a very great extent, 18.9 % to a great extent, 21.1% to a moderately, 40% to slightly, and 15.6% to rarely. And only 55.1% of the respondents had

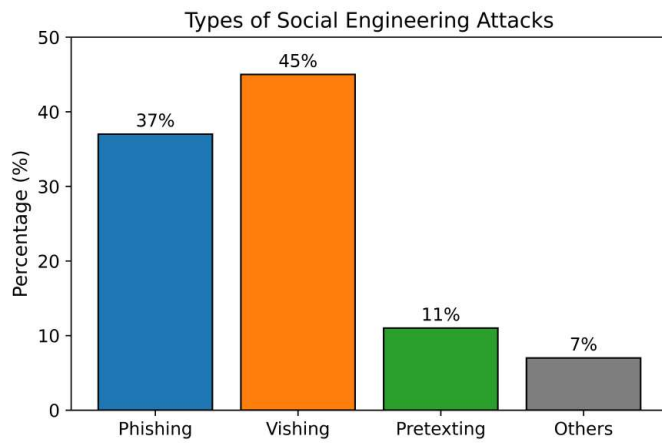


Fig. 5. Comparison of social engineering attacks.

not experienced a vishing attack at all. Measurements of the attacks for vishing are given in Figure 6 by the bar chart.

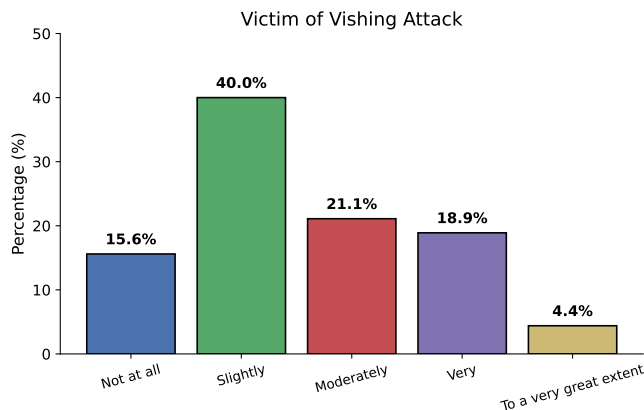


Fig. 6. Measurements of the attacks for vishing.

C. Revelation to Vishing Attacks caused by Technical Factors

The research aimed to find technical factors related to mobile phone use that make users in the country more vulnerable to phishing attacks. Maximum of the users 21.1% have reported that their phones had very little complexity, 38.9% had slightly, 16.7% had moderately, 20% had more complexity and only 3.3% had much complexity on their mobile phones, shown in Figure 7. As a result, they were vulnerable to vishing assaults due to the intricacy of their mobile devices use.

D. Exposure due to human factors

Instead of systems with security mechanisms in place, social engineers favor compromising people by manipulating their psyche. Figure 8 is showing the majority of the respondents 35.6% attested that their psychological factor had little potential, whereas only 11.1% attested that their psychological matters had a noteworthy and critical potential respectively in exposure to vishing attacks.

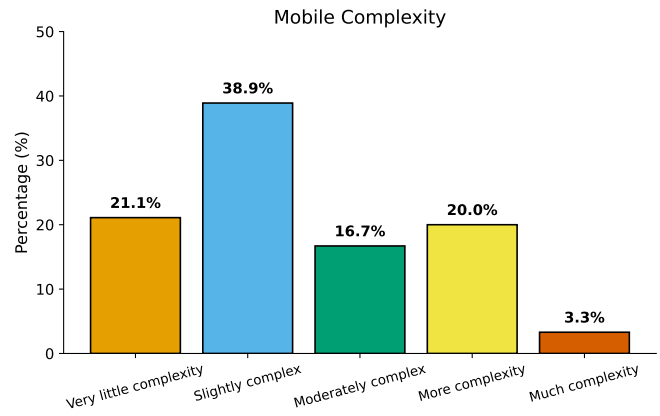


Fig. 7. Complexity of using the mobile phone.

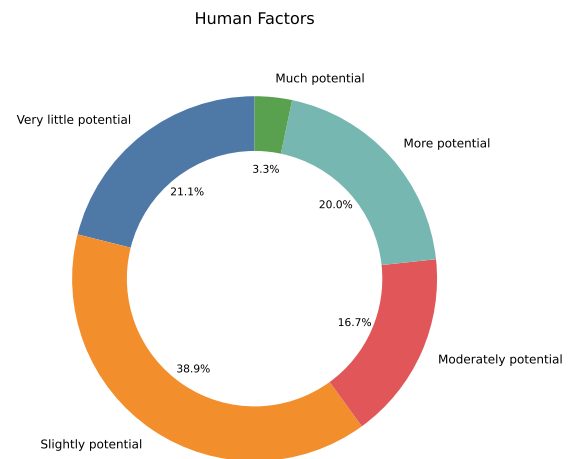


Fig. 8. Exposure due to human factors.

E. Subjection to Vishing Attack Due to MFS Services

This study aims to find the domination of vishing attacks due to MFS services. Study shows at the Figure 9 which is the majority of the 35.6% experienced vishing attacks due to MFS services to a very great extent and only 12.2% of the respondents had experienced them to a very little extent.

F. The consciousness of Vishing Attack

The study sought to find out the consciousness at Figure 10 of vishing attacks through the responders. Nearly 81% of respondents had very little knowledge of vishing attacks and their simple prevention strategies whereas the remaining numbers responded as familiar with this type of criminal activity.

G. Safeguard mechanisms used by User

The study aims to find out the safeguard mechanism in Figure 11, which is used by the user to protect them from vishing attacks. Approximately four-fifths of respondents use almost no safeguard mechanism, while 6.7% of them use it as a protective pathway. After analyzing the data given by the

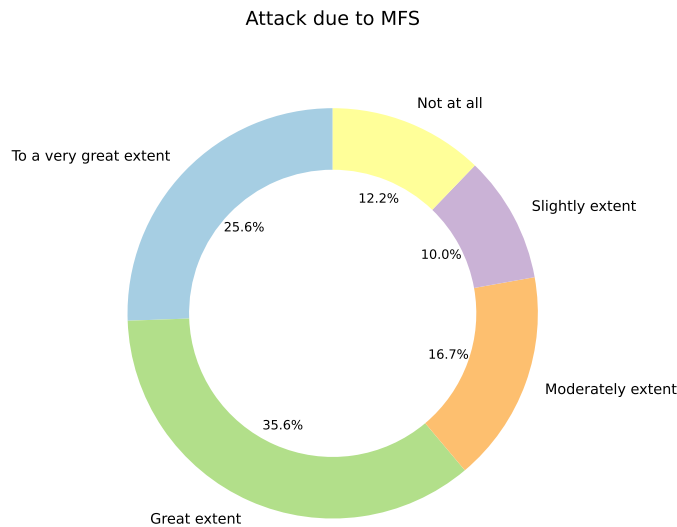


Fig. 9. Vishing attacks associated with Mobile Financial Services (MFS).

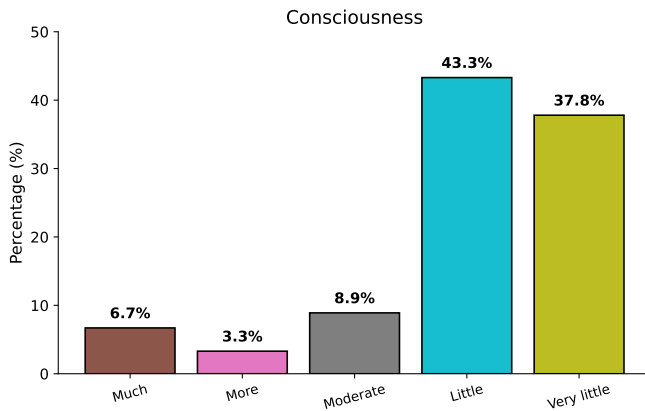


Fig. 10. Level of consciousness about vishing attacks among respondents.

end user, we conclude that mobile complexity, psychological factors as well as very little knowledge about the security system, ensure an increase in vishing attacks on a larger scale. It is clear from the survey that the degree of knowledge needed to alleviate the vishing attack is not sufficient to secure the MFS account of an individual.

IV. CONCLUSION

The suggested strategy for detecting phishing attacks is workable for defending social engineering attacks. By dividing the decision-making process into manageable parts, this model helps in the detection of social engineering assaults, which is a significant addition to the field of social engineering. Empirical testing using real vishing-call audio remains future work due to dataset unavailability. Users of mobile devices can use the model as a weapon to defend themselves against all types of vishing attacks.

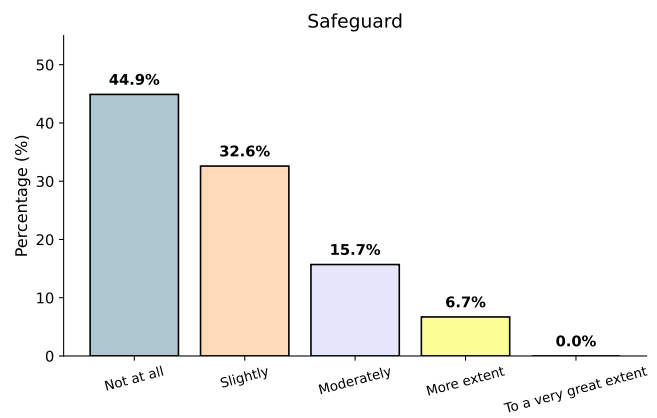


Fig. 11. Safeguards adopted by users to protect against vishing attacks.

REFERENCES

- [1] I. Veitaitė, "Challenges of cyber security in modern society: The impact of social engineering," *Scientific and practical cyber security journal*, 2024.
- [2] H. J. Akeiber, "The evolution of social engineering attacks: A cyber-security engineering perspective," *Al-Rafidain Journal of Engineering Sciences*, pp. 294–316, 2025.
- [3] M. Akhtaruzzaman, M. E. Islam, M. S. Islam, S. Islam, M. R. U. Bhuiyan, M. Tareq, and M. J. U. Palas, "An impact study on mobile financial services (mfss) in bangladesh," *Department of Communications and Publications, Bangladesh Bank. Retrieved September*, vol. 27, p. 2020, 2017.
- [4] B. Bank *et al.*, "Mobile financial services in bangladesh: An overview of market development," *Bangladesh Bank Policy Paper*, pp. 1–28, 2012.
- [5] M. N. Nabi and M. T. Islam, "Cyber security in the globalized world: Challenges for bangladesh," *Economic and Social Development: Book of Proceedings*, p. 32, 2014.
- [6] M. S. R. Bhuiyan and M. M. Rahman, "Implementation of mobile banking in bangladesh: opportunities and challenges," *IOSR Journal of Electronics Communication Engineering*, vol. 15, no. 4, pp. 53–58, 2013.
- [7] C. F. M. Foozy, R. Ahmad, M. F. Abdollah, R. Yusof, and M. Z. Mas'ud, "Generic taxonomy of social engineering attack and defence mechanism for handheld computer study," in *Malaysian Technical Universities International Conference on Engineering & Technology, Batu Pahat, Johor*. sn, 2011.
- [8] K. S. Jones, M. E. Armstrong, M. K. Tornblad, and A. Siami Namin, "How social engineers use persuasion principles during vishing attacks," *Information & Computer Security*, vol. 29, no. 2, pp. 314–331, 2021.
- [9] A. Qamar, A. Karim, and V. Chang, "Mobile malware attacks: Review, taxonomy & future directions," *Future Generation Computer Systems*, vol. 97, pp. 887–909, 2019.
- [10] P. P. Parthy and G. Rajendran, "Identification and prevention of social engineering attacks on an enterprise," in *2019 International Carnahan Conference on Security Technology (ICCSST)*. IEEE, 2019, pp. 1–5.
- [11] E. O. Yeboah-Boateng and P. M. Amanor, "Phishing, smishing & vishing: an assessment of threats against mobile devices," *Journal of Emerging Trends in Computing and Information Sciences*, vol. 5, no. 4, pp. 297–307, 2014.
- [12] K. Choi, J.-I. Lee, and Y.-t. Chun, "Voice phishing fraud and its modus operandi," *Security Journal*, vol. 30, no. 2, pp. 454–466, 2017.
- [13] M. Bezuidenhout, F. Mouton, and H. S. Venter, "Social engineering attack detection model: Seadm," in *2010 Information Security for South Africa*. IEEE, 2010, pp. 1–8.